

Requirements Specification

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DOCUMENT HISTORY

Version	Date	Changes made	Sign	Reviewer
0.1	2025-09-08	First draft submitted for review	AF, AG, VG, MM	DA
0.2	2025-09-19	Revised draft after feedback; requirements rewritten to be modular and measurable; updated Use section, group info, and document history	AF, AG, VG, MM	DA
0.3	2025-09-23	Revised per feedback; measurable performance reqs; Use section	AF, AG, VG, MM	DA
1.0	2025-11-07	Approved by supervisors (Tollgate 1). Basis for design and implementation.	AF, AG, VG, MM	DN

1 INTRODUCTION

1.1 Project Information

The Music Transcription project develops a system that converts keyboard performances into a sequence of notes or a note sheet. Using a Casio CTK-3500 keyboard and a Streamplify MIC ARM microphone, along with the appropriate USB cabling, the system will capture audio from live playing or pre-recorded input. It must detect the onset and offset of each note, identify the fundamental frequency and harmonics, estimate the note duration, and map these measurements to the correct pitch. The system will support both microphone-based recording and MIDI input; a MIDI file obtained via the cable can serve as a reference signal. Signal-processing techniques and machine-learning algorithms will be combined to implement pitch detection, onset/offset detection, and automatic transcription, producing an output such as “C4 C4 A5” or a visual note sheet.

1.2 Partners

The Music Transcription project is carried out by Group 3 in the CDIO project course TSKS23. Our team comprises four second-year master’s students at Linköping University:

- **Project Lead:** Ahmed Fawaizat(fawah595@student.liu.se)
- **Design Lead:** Abdul Gaffar Abubakar Passum (abuab954@student.liu.se)
- **Test Lead:** Varun Gurupurandar (vargu125@student.liu.se)
- **Documentation Lead:** Mehran Mamivand(mehma172@student.liu.se)

We are undertaking this project for the Division of Communication Systems at Linköping University. The customer and sponsor is Dr Danyo Danev(danyo.danev@liu.se), who is the examiner for the course and will review and approve our design and final deliverables.

In addition to our team and the customer, the project involves supervisors appointed by the department, David Nordlund (david.nordlund@liu.se), Oksana Moryakova (oksana.moryakova@liu.se), and Henrik Åkesson (henrik.akesson@liu.se), who provide technical guidance and feedback on our work. Collectively, these partners ensure that our work meets the academic, technical, and project-management standards of the CDIO course.

1.3 Use

The product is intended as an automatic transcription tool that converts live or recorded piano input into a digital representation of notes. Its primary purpose is to assist musicians, educators, and researchers by providing a reliable transcription aid for practice, analysis, and documentation. The system supports both microphone-based recordings and pre-recorded audio or MIDI input, and outputs the results as note lists, MIDI files, or simple sheet notation.

The initial system is designed as a stand-alone prototype for demonstration and evaluation. Optional real-time functionality may be explored, but it is considered a future extension and not part of the baseline version.

The graphical user interface (GUI) shall enable users to:

- Start and stop recording from the microphone and save the audio as a .wav file.
- Load a pre-recorded .wav or .midi file for offline transcription.
- Process the input signal to detect note onset, offset, pitch, and duration, and display the detected notes with timing information.
- Export the results as a .midi file or a simple note sheet.

1.4 Background information

Automatic music transcription (AMT) has attracted increasing attention because it promises to convert raw audio into symbolic music representations automatically. Advances in signal processing and machine learning have

greatly improved transcription accuracy, yet existing systems still struggle to match human experts. Developing a working AMT system, therefore, offers a valuable opportunity for students to apply theoretical knowledge to a challenging real-world problem, explore the limitations of current algorithms, and contribute to ongoing research in this field.

Piano signals are well-suited for such a study because they are approximately periodic and can be represented as a Fourier series consisting of a fundamental frequency (f_0) and its harmonics. Western music uses a seven-note alphabet (A, B, C, D, E, F, G) repeated across octaves, with each octave divided into twelve semitones. Our 61-key Casio CTK-3500 keyboard spans five octaves; its tones are recorded either via a microphone or as MIDI data.

This project aims to implement and evaluate classifier algorithms that take an audio signal from the keyboard and determine what notes were played. Because AMT encompasses pitch detection, onset/offset detection, and duration estimation, we divide the work into mini-projects: single-pitch identification, dual-pitch (chord) identification, detection of multiple notes and their durations, transcription of simple pieces, and more complex scenarios. By working through these tasks, students gain hands-on experience with signal processing, machine learning, system design, and evaluation, while addressing an open problem in music technology.

1.5 Purpose of the Requirement Specification

This document formalizes the agreed-upon features of the music-transcription system. It serves as the basis for design, planning, and evaluation. The customer will review and approve the specification before detailed design begins.

1.6 Definitions

1.6.1 Project-related definitions

- **Pitch:** The perceived frequency of a note, associated with the fundamental frequency f_0 of a periodic musical tone.
- **AMT (Automatic Music Transcription):** The process of analyzing an acoustic musical signal to determine pitches, onset times, durations, and sources of sounds.
- **MIDI (Musical Instrument Digital Interface):** A standard for exchanging performance data between electronic musical devices; stores note information (e.g., which notes to play) but not audio waveforms.
- **Fundamental Frequency (f_0):** The lowest frequency in a note's Fourier expansion.
- **Harmonics (Overtones):** Integer multiples of the fundamental frequency (e.g., $2f_0$, $3f_0$).
- **Octave:** An interval where the frequency doubles; divided into 12 semitones.
- **Onset:** The instant when a note begins, marked by a rise in amplitude.
- **Offset:** The end of a note's duration.
- **Spectrogram:** A visual representation of the signal's frequency content over time.
- **LIPS Project Model:** A framework for project management, including tollgates, deliverables, and phases (before, during, after).
- **Tollgate:** A checkpoint meeting with the customer to approve progression to the next project phase.

1.6.2 Requirement Definitions

- **Priority 1:** Basic requirement, the requirement must be met at the delivery.
- **Priority 2:** Extra requirement, the requirement must be fulfilled if time is left in the budget when all requirements with priority 1 are fulfilled.
- **Priority 3:** These requirements are only to be worked on if time is left in the budget when every other requirement is fulfilled. They can also be seen as development opportunities if the system were to be further developed.

Structure of requirements list:

Requirement	Version	Description	Priority
Number	Original or changed, if so which iteration?	Description of requirement	1/2/3
Number	Original or changed, if so which iteration?	Description of requirement	1/2/3

2 REQUIREMENTS

2.1 System Requirements

Requirement Number	Version	Description	Priority
1	1.0	Transcribe music signals recorded via microphone from the Casio CTK-3500 keyboard into a basic list of notes (e.g., C4, C5, A5) using signal processing techniques.	1
2	1.0	Implement pitch detection using signal processing (e.g., Fourier analysis) to identify fundamental frequencies and harmonics.	1
3	1.0	Implement onset detection using signal processing (e.g., energy increases and spectral changes) to determine note start times.	1
4	1.0	Implement offset detection using signal processing (e.g., energy decreases and spectral changes) to determine note end times.	1
5	1.0	Integrate automatic transcription logic combining pitch and onset/offset data via signal processing.	1
6	1.0	Handle single pitches, identifying one note at a time including its duration via signal processing.	1
7	1.0	Handle multiple pitches (up to 2 notes simultaneously) for simple pieces, including durations via signal processing.	1
8	1.0	Interface with Casio CTK-3500 keyboard via USB-B for MIDI input.	1
9	1.0	Interface with MIC ARM microphone via USB-C to USB-A/C for audio capture.	1
10	1.0	Connect and process signals on a laptop.	1
11	1.0	Use MIDI files obtained via cable as reference signals (ground truth) for validation and accuracy comparison.	1
12	1.0	Analyze pre-recorded audio signals (e.g., WAV files).	1
13	2.0	Analyze live audio signals in non-real-time mode.	1
14	1.0	Use Python (with libraries like PyAudio and Wave) for audio recording and signal processing.	1
15	2.0	Collect and analyze large music samples to improve signal processing for applications.	2

16	2.0	Optionally incorporate machine learning algorithms to enhance pitch, onset/offset, and transcription (low focus).	3
17	2.0	Support real-time transcription, updating the note list after each note is played using signal processing.	3

2.2 User Interface Requirements

Requirement Number	Version	Description	Priority
1	1.0	Present output as a text list of notes (e.g., "C4 C5 A5").	1
2	1.0	Allow users to upload recorded WAV files for processing.	1
3	1.0	Support live recording with options to pause and play the recording.	1
4	1.0	Provide input options for selecting recording mode (microphone or cable).	1
5	1.0	Allow selection of recording duration.	1
6	1.0	Include options for setting signal processing parameters (e.g., frequency thresholds, sensitivity for detection).	1
7	1.0	Display detected onsets and offsets visually (e.g., markers on waveforms or spectrograms).	1
8	2.0	Display plots such as spectrograms for the analyzed signal.	2
9	2.0	Show comparison of detected notes to ground truth (e.g., from MIDI reference).	2
10	2.0	Provide log files or on-screen log information for processing details.	2
11	2.0	Present output as a visual note sheet or MIDI track.	2
12	2.0	Allow users to record audio/MIDI inputs through the interface.	2
13	2.0	Process the recorded or uploaded signal and display the transcription.	2
14	2.0	Include a user manual describing equipment setup, signal recording, and output interpretation.	2
15	2.0	Dynamically update the transcription in real-time mode as notes are played.	3

2.3 Performance Requirements

Requirement Number	Version	Description	Priority
1	1.0	The system shall correctly identify at least 80% of single-note pitches in test recordings. Verification will be performed using annotated single-note datasets to compute pitch accuracy.	1
2	1.0	The system shall correctly identify at least 60% of pitches in recordings containing two simultaneously played notes (polyphonic detection). Verification will be done using annotated chord recordings and pitch accuracy computation.	2
3	1.0	The system shall process a 30-second input audio file within 10 seconds on a standard laptop (Intel i5, 8 GB RAM). Verification will be performed by timing the processing of benchmark recordings.	2

4	1.0	The system shall detect note onsets such that the mean absolute error between detected and reference onset times does not exceed 50 ms . Verification will be performed by comparing estimated onset times against annotated ground truth timestamps.	1
5	1.0	The system shall detect note offsets such that the mean absolute error between detected and reference offset times does not exceed 70 ms . Verification will be performed by comparing estimated offset times against annotated ground truth timestamps.	1
6	1.0	The system shall estimate note durations with a mean absolute error not exceeding 0.1 s , calculated as the absolute difference between detected and annotated note durations.	1

2.4 Economy

- The project must operate within allocated resources: 15 hours of supervision, 25 hours of expert consultancy, and access to the communications lab at ISY and its computers.
- No additional budget is specified; existing equipment (keyboard, microphone, cables, laptop) must be used.
- Software must be free or cross-platform (e.g., Python, MATLAB, Cakewalk by BandLab), with no installations requiring internet or additional packages beyond provided libraries.

2.5 Delivery Deadlines

Item	Description	Target Date
Tollgate 2	Requirement Specification, System Design Sketch, Project Plan	September 22, 2025 (three weeks from September 1)
Tollgate 3	Design Specification, Test Plan	October 20, 2025 (aligned with mid-project progression)
Tollgate 5	Test Protocols, User Manuals, demonstration of complete functionality	November 24, 2025 (end of during-phase)
Tollgate 6	Technical Report, Post-Project Review, Poster Presentation, Oral Presentation, Project Web Page	December 15, 2025 (before project conference)
Weekly Reports	Progress description, individual/collective time reports, revised time plan	Every Monday at 12:00, starting September 29, 2025
Final Product	All deliverables, including final product	December 15, 2025 (before project conference)

2.6 Notes

- **Post-Project Review:** May be updated within one week post-Tollgate 6 (by December 22, 2025).